AI-Powered Workflow Automation

Course Module & Curriculum Plan

Target Audience: Students, beginners, or professionals looking to streamline operations using automation and AI tools.
Prerequisites: Basic computer literacy (no coding experience required).

2. Topics

Topic 1: Introduction to Automation

What is Automation? Automation is the process of performing repetitive tasks automatically with minimal manual effort.

Real-life examples:

- Sending an email after someone submits a form.
- Saving form responses in a spreadsheet.
- Generating a report from collected data.
- Sending a reminder before an important deadline.
- Categorizing customer messages automatically.

Topic 2: Automation vs. AI Automation

Traditional automation: Follows predefined rules and instructions.

Example: When a form is submitted, send a confirmation email.

AI automation: Uses AI to interpret or generate information as part of a workflow.

Example: Read a customer complaint, identify its category, summarize the issue, and prepare a suitable response for approval.

Topic 3: Important Components

1. **Trigger:** Starts the workflow.
2. **Input:** Provides the information to process.
3. **Action:** Performs an operation.
4. **Condition:** Determines which path to follow.
5. **AI Processing:** Generates, classifies, summarizes, or extracts information.
6. **Output:** Produces the result.
7. **Error Handling:** Manages failures or unexpected results.

Example Workflow (Student Contact Form):

Form Submission → Validate Input → **AI Categorizes Message** → Save in Spreadsheet → Prepare Email → **Human Review**

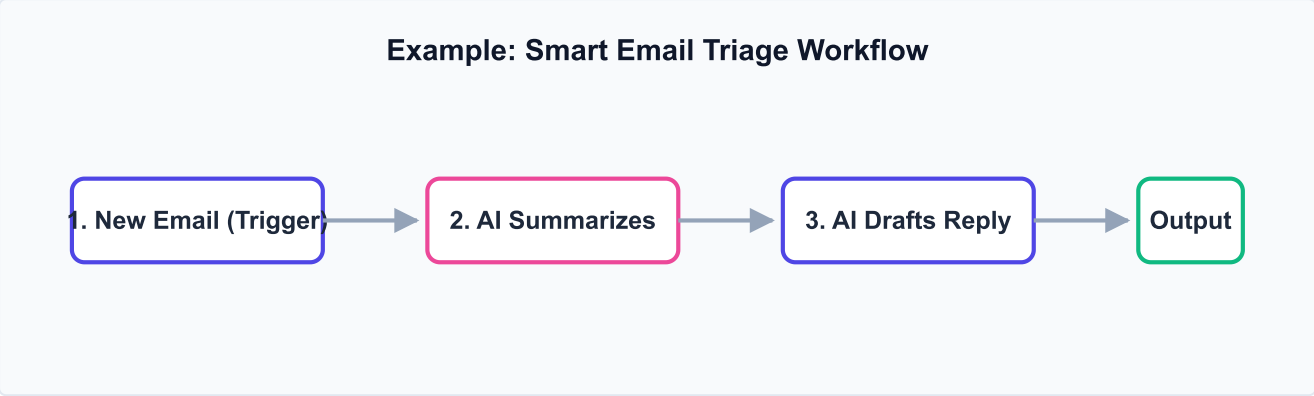
Lesson 3: The Automation Toolkit & Webhooks

Objective: Use no-code and low-code automation tools; Understand webhooks and how applications exchange data.

- **Introduction to No-Code Platforms:** Overview of industry standards like Zapier, Make (Integromat), or Microsoft Power Automate. Navigating the interface, connecting accounts, and testing steps.
- **How Apps Talk to Each Other:**
  - **APIs (Application Programming Interfaces):** The "waiter" taking a request to the kitchen and bringing the data back.
  - **Webhooks:** The "doorbell". A method for an app to instantly push data to another app as soon as an event happens, eliminating the need to constantly ask for updates.
- **Practical Exercise:** Setting up a simple webhook to catch data from a free webhook tester and route it to a spreadsheet.

Lesson 4: AI Integration & Practical Workflows

Objective: Integrate AI models with applications and APIs; Build workflows for email, forms, reports, and student management.



Workflow Build: Form-to-Report Automation
When a user submits a form, an AI model automatically categorizes the submission based on predefined instructions, updates a CRM database, and generates a personalized PDF report that is automatically emailed to the submitter.

Lesson 5: Governance, Error Handling & HITL

Objective: Add error handling, verification, and human approval to automated workflows.

- **Designing for Failure (Error Handling):** What happens when an API is down or a step fails? Setting up error routes (e.g., "If step 3 fails, send an alert email to the admin instead of stopping completely").
- **The "Human-in-the-Loop" (HITL) Approach:** Why AI shouldn't act entirely autonomously in sensitive situations. Creating workflows that pause and wait for a human to click "Approve" or "Reject" via Slack or Email before executing a final action.
- **Data Verification:** Ensuring data formatting (dates, phone numbers) is clean before it hits your database.

Lesson 6: Capstone Project

Objective: Design a small AI-powered automation project solving a real-world problem.

Students must design, build, and test a multi-step automation workflow. The deliverable is a live demonstration of the workflow executing successfully, accompanied by a flowchart diagram explaining the logic.

Requirement	Description	Points
Structure	Must use at least one Trigger and two distinct Actions.	25 pts
Logic	Must include at least one Condition/Filter branch.	25 pts
AI Integration	Must integrate an AI prompt step for dynamic generation or analysis.	25 pts
Governance	Must include a Human Approval (HITL) step before final output.	25 pts

3. Practical Exercises

Practical 31: Automated Form-to-Sheet Workflow

Task: Create a student registration form and store submissions in Google Sheets.

Students should:

1. Create fields for name, email, and course.
2. Collect sample submissions.
3. Verify that responses are saved correctly.
4. Identify missing or duplicate entries.

Learning: Data collection and workflow automation.

Practical 32: AI Email Draft Generator

4. Classroom Challenge

Build an AI-Powered Student Support Workflow

Task: Create a comprehensive workflow that automatically processes and routes student queries.

Requirements:

1. Accept a query through a form.
2. Validate the submitted data.
3. Categorize the query using AI.
4. Prepare a short summary.
5. Route the query according to its category.
6. Save the query and result in a spreadsheet or database.
7. Flag urgent or unclear queries for human review.
8. Record failures and test the workflow using sample inputs.

Learning Outcome: Students will demonstrate a complete understanding of triggers, data validation, AI integration, conditional routing, and human-in-the-loop exception handling in a single, unified project.

5. Standard AI Automation Formula

Trigger + Input + AI Processing + Conditions + Actions + Verification + Output

Example:

A customer submits a complaint → Validate the message → AI categorizes the issue → Apply routing rules → Prepare a response → Request approval where necessary → Save the result.

6. Homework / Assignment

1. Identify five repetitive tasks in education or business that could be automated.
2. Draw a workflow diagram for one selected task.
3. Create a prompt that produces structured output for automation.
4. Explain the difference between an API and a webhook.